

WASP-2b

R-2071

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_121012 · created 2026-08-09T23:14:27+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456212.6044 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.047
Transit depth (Rp/Rs)^2	2.36 +/- 1.43 [%]
Orbital Inclination (inc)	82.86 +/- 2.84
Airmass coefficient 1 (a1)	0.34 +/- 0.17
Airmass coefficient 2 (a2)	1.13 +/- 1.6
Scatter in the residuals of the lightcurve fit is	46.96 %
Best Comparison Star	None
Optimal Aperture	0.89
Optimal Annulus	3.85
Transit Duration (day)	0.022 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.047 vs 0.1326 (deviation 0.45σ)
Duration fit vs published	0.022 d vs 0.0737 d (deviation 2.72σ)
Mid-time O–C	-5.9 min
Depth SNR	3.3
Residual scatter	46.96 %

Light curve

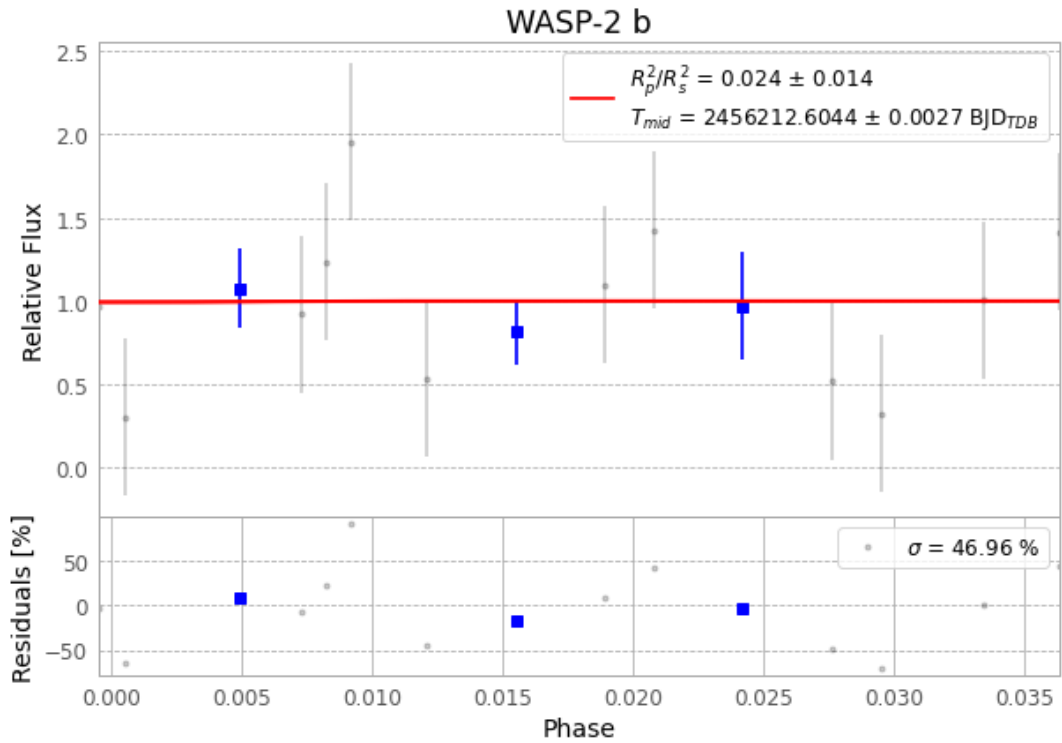


Figure 1 — FinalLightCurve_WASP-2 b_2012-10-11.png (SHA-256 48382996c102f0b2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [58, 190], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 01927f6a8dede1f7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

43 FITS frames (28 MB), WASP-2121012022113.FITS → WASP-2121012043612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-121012.log (d983ba5a37aa...), FinalLightCurve_WASP-2 b_2012-10-11.png (48382996c102...), FinalLightCurve_WASP-2 b_2012-10-11.csv (92802a4b2ecb...)

Compute resources

Wall time 197.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 23:14Z.